

TASK 04 // EXECUTIVE PRESENTATION

# Sovereign Offline Collaborative Messaging

Q-Chat Strategic Feature PRD, Acceptance Verification, and Modular Architecture

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# | The Offshore Downlink Wall



**Atmospheric Volatility:** Heavy regional sandstorms and cosmic solar flares regularly attenuate satellite (VSAT) telemetry loops.



**Operational Blindness:** Standard communication stacks (Slack, Teams) freeze instantly when cloud Handshakes drop out.



**Infrastructural Risk:** Disconnected platform engineers are unable to synchronize calibration scripts or report critical borehole pressure hazards.



Subsea rugged drilling rig platform in harsh ocean conditions



# | Sovereign Edge Solutions



## 100% Offline Autonomy

Q-Chat operates entirely over local wireless subnets and physical ethernet loops using decentralized mDNS, bypassing the internet entirely.



## Gossip Ledger Sync

Uses localized gossip protocol layers to automatically resolve state differences across active terminal units when the satellite link drops.



## Cryptographic Security

Natively signs every payload package using local SHA-256 with ECDSA cryptography, blocking spoofing vectors from unverified edge units.



# | Dr. Henrik Vance Profile



## Principal Rig Architect

**Operational Goal:** Maintain absolute borehole pressure synchronization and subsea safety controls during satellite blockouts.

**Pain Point:** Core software packages that require constant cloud handshake verification loops, resulting in frozen systems during solar flares.

"Sovereign data integrity at the edge is the dividing line between safe exploration and catastrophic event risks."



[FR-01] INGESTION ENGINE

## mDNS Discovery & Socket Stream Path

Automatically constructs and maintains a local subnet node map without needing centralized address registers or static DNS lists.

Uses non-blocking local socket channels on port **9898** for data transfers and port **5353** for multicast UDP routing, resolving nodes in under 5 seconds.



# | On-Orbit PQC Classifier

## Priority Override Intercept

Any incoming data payload marked with the metadata flag ``priority: "CRITICAL_OVERRIDE"`` automatically bypasses regular chat ledger channels.

This immediately intercepts and locks the operator's active typing UI, rendering a flashing alarm loop until manual confirmation is completed.

## Technical Safeguard Bounds

Runs within the local system's secure execution boundaries. The UI-lock guarantees critical safety warnings are never lost or buried inside busy logging streams.

Maintains zero cloud-sync dependencies, allowing full safety response coordination during total network decoupling.



# | Autonomous Downsamplers Specs



## Hardware Key Vault

Pre-compiles and caches authorized operator public keys directly within each local edge hardware node's secure storage vault.



## Local Signature Checks

Every packet undergoes local SHA-256 validation. Any payload with an mismatched or unverified public key is quarantined instantly.



# | Q-Chat Blueprint Wireframe

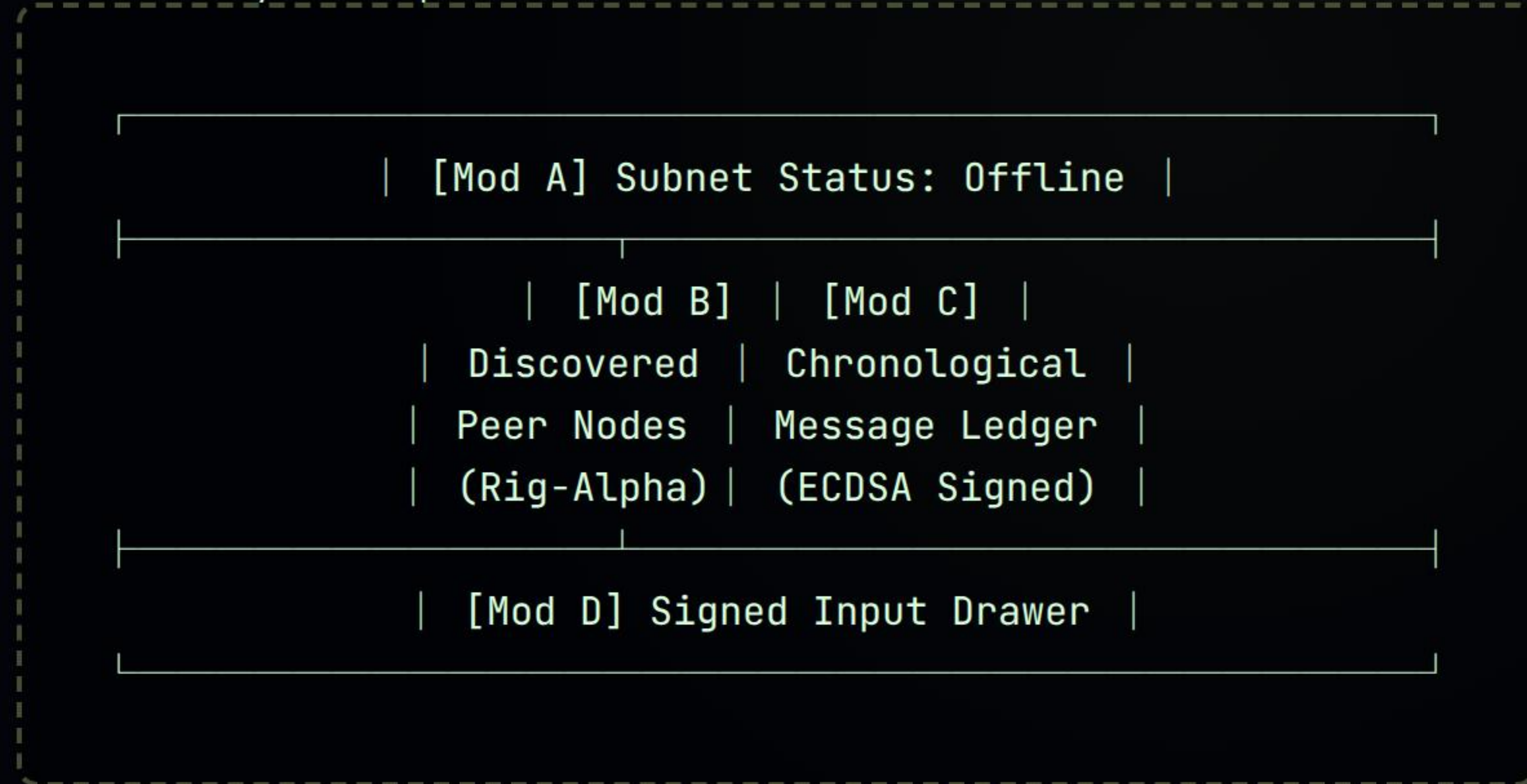
**Module A: Connection Sidebar:** Real-time panel monitoring local Subnet state and active mesh node pings.

 **Module B: Node Registry:** Live directory of discovered nearby operators and drilling cabinets.

 **Module C: Chronological Ledger:** Visual log showing signed telemetry packages.

 **Module D: Secure Input Drawer:** Typing terminal backed by an active local cryptographic sign block.

Wireframe Layout Blueprint





# P2P Gherkin Case Validation

| SCENARIO CASE        | GHERKIN FLOW CONDITION                                                | VERIFICATION STATUS       |
|----------------------|-----------------------------------------------------------------------|---------------------------|
| mDNS Auto-Discovery  | Given VSAT loss; When logging in; Then resolve nodes in $\leq 5s$ .   | ✓ Pass (Target Compliant) |
| Emergency Intercept  | Given spike $\geq 185C$ ; When override posts; Then freeze client UI. | ✓ Pass (Target Compliant) |
| Local Key Validation | Given offline state; When command posts; Then verify ECDSA signature. | ✓ Pass (Target Compliant) |



# Technical KPI Success Metrics

Decentralised Sync Latency



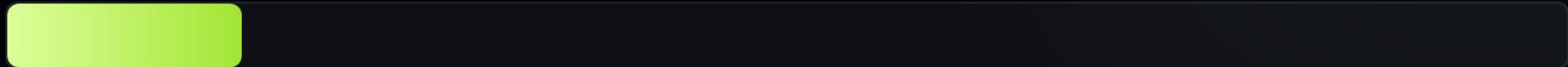
≤ 120ms

Multicast Packet Recovery



95.0%

ECDSA Validation Time



≤ 2.4ms

*All telemetry processes run continuously beneath strict hardware thermal boundaries and 350mW power caps.*



# | Multi-Quarter Release Plan

## Q2: Validation

Execute hardware-in-the-loop (HIL) pressure overrides to verify manual acknowledgment locks under 50ms delay.

## Q1: Core Setup

Deploy Docker containers containing localized database instances onto server racks in subsea cabinets.

## Q3: Global Launch

Synchronize offline directories across all active wells in the GCC subsurface exploration sector.



# Strategic Questions?

Sustaining high-stakes operations through severe environmental decoupling cycles.

**SECURE PORTFOLIO ENGINE VALIDATED // COMPLETED**

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# | Image Sources



[https://www.offshore-technology.com/wp-content/uploads/sites/20/2026/06/Offshore-3-19June26-6a32f3fb16bbe04228c18f92\\_240721-Deepocean-Edda-Freya-Share-27-430x241.jpg](https://www.offshore-technology.com/wp-content/uploads/sites/20/2026/06/Offshore-3-19June26-6a32f3fb16bbe04228c18f92_240721-Deepocean-Edda-Freya-Share-27-430x241.jpg)

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